PRACTICAL 7

AIM: React Front-end:

- Create ProductList, ProductDetail, Cart, Checkout components.
- Use React Router for page navigation.
- Integrate APIs with Axios & manage global state using Context.

Create APP

```
PS C:\Users\16raj\OneDrive\Desktop\SEM VI> npx create-react-app ecommerce-app
Need to install the following packages:
create-react-app@5.1.0
Ok to proceed? (y) Y
```

Create following structure

```
src/
 ├── components/
 │    ├── ProductList.js
 │    ├── ProductDetail.js
 │    ├── Cart.js
 │    └── Checkout.js
 │
 ├── context/
 │    └── CartContext.js
 │
 ├── services/
 │    └── api.js
 │
 ├── App.js
 └── index.js
```

Install packages

```
PS C:\Users\16raj\OneDrive\Desktop\SEM VI\ecommerce-app> npm install react-router-dom axios

added 7 packages, and audited 1320 packages in 6s

268 packages are looking for funding
  run `npm fund` for details
```

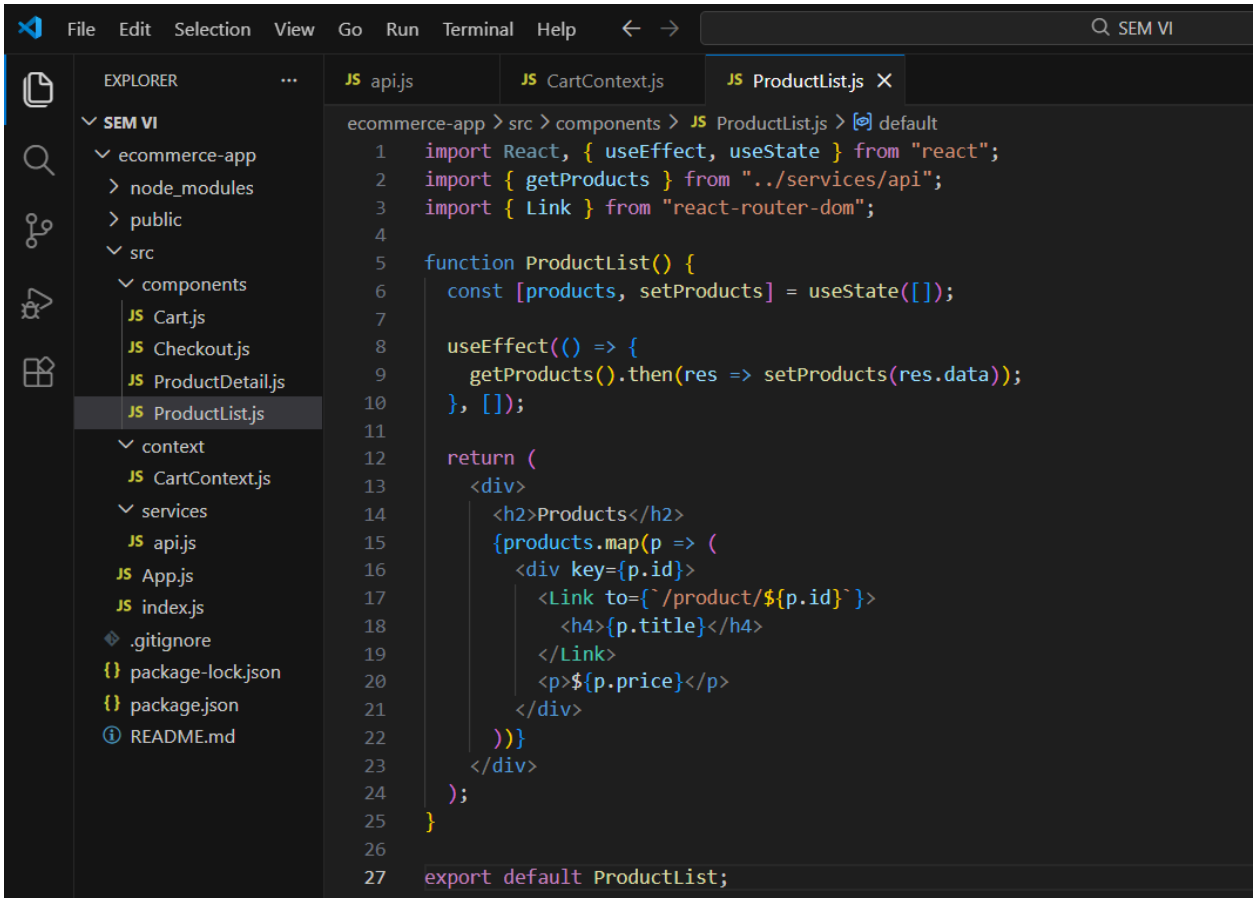
API integration using AXIOS - services/[api.js]

```
ecommerce-app > src > services > api.js > ...
import axios from "axios";

const API = axios.create({
  baseURL: "https://fakestoreapi.com"
});

export const getProducts = () => API.get("/products");
export const getProduct = (id) => API.get(`/products/${id}`);
```

Global State using Context - context/[CartContext.js]

```
ecommerce-app > src > context > CartContext.js > ...

export const CartProvider = ({ children }) => {
  const [cart, setCart] = useState([]);

  const addToCart = (product) => {
    setCart([...cart, product]);
  };

  const removeFromCart = (id) => {
    setCart(cart.filter(item => item.id !== id));
  };

  return (
    <CartContext.Provider value={{ cart, addToCart, removeFromCart }}>
      {children}
    </CartContext.Provider>
  );
};
```

Product List Component - components/ProductList.js



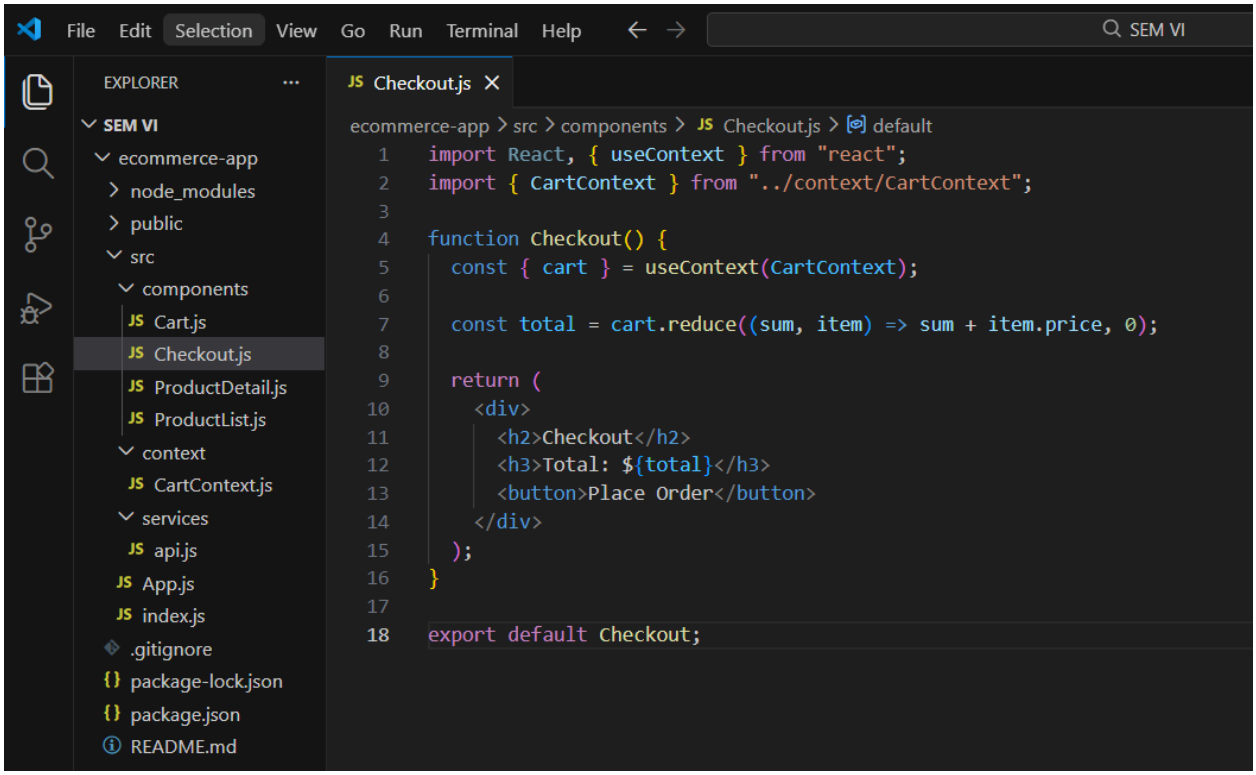
Product Detail Component - components/[ProductDetail.js]

```
ecommerce-app > src > components > ProductDetail.js > ProductDetail
import React, { useEffect, useState, useContext } from "react";
import { useParams } from "react-router-dom";
import { getProduct } from "../services/api";
import { CartContext } from "../context/CartContext";

function ProductDetail() {
  const { id } = useParams();
  const [product, setProduct] = useState({});
  const { addToCart } = useContext(CartContext);

  useEffect(() => {
    getProduct(id).then(res => setProduct(res.data));
  }, [id]);

  return (
    <div>
      <h2>{product.title}</h2>
      <p>{product.description}</p>
      <h3>${product.price}</h3>
      <button onClick={() => addToCart(product)}>
        Add to Cart
      </button>
    </div>
  );
}

export default ProductDetail;
```

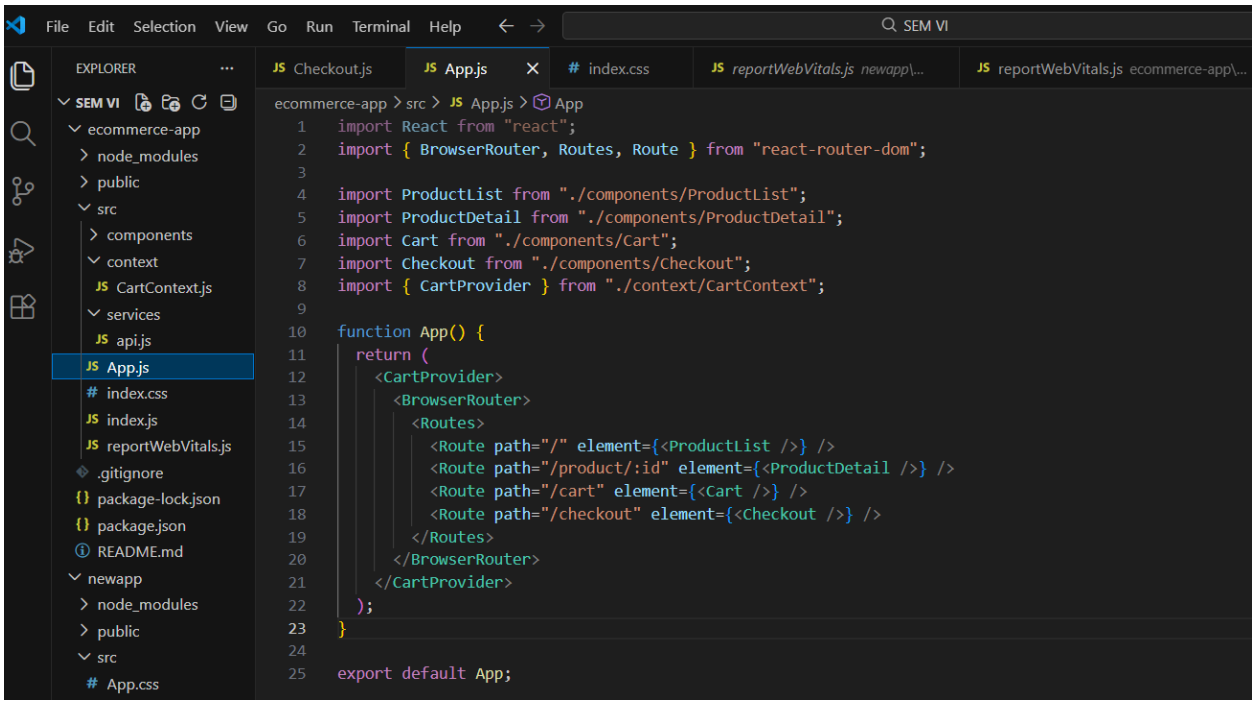
Cart component - components/[Cart.js]

```
ecommerce-app > src > components > Cart.js > default
import React, { useContext } from "react";
import { CartContext } from "../context/CartContext";

function Cart() {
  const { cart, removeFromCart } = useContext(CartContext);

  return (
    <div>
      <h2>Cart</h2>
      {cart.map(item => (
        <div key={item.id}>
          <h4>{item.title}</h4>
          <button onClick={() => removeFromCart(item.id)}>
            Remove
          </button>
        </div>
      ))}
    </div>
  );
}

export default Cart;
```

Checkout Components - components/[Checkout.js]



Routing setup - [App.js]



OUTPUT

localhost:3000

Products

Fjallraven - Foldsack No. 1 Backpack, Fits 15 Laptops

$109.95

Mens Casual Premium Slim Fit T-Shirts

$22.3

Mens Cotton Jacket

$55.99

Mens Casual Slim Fit

$15.99

John Hardy Women's Legends Naga Gold & Silver Dragon Station Chain Bracelet

$695

Solid Gold Petite Micropave

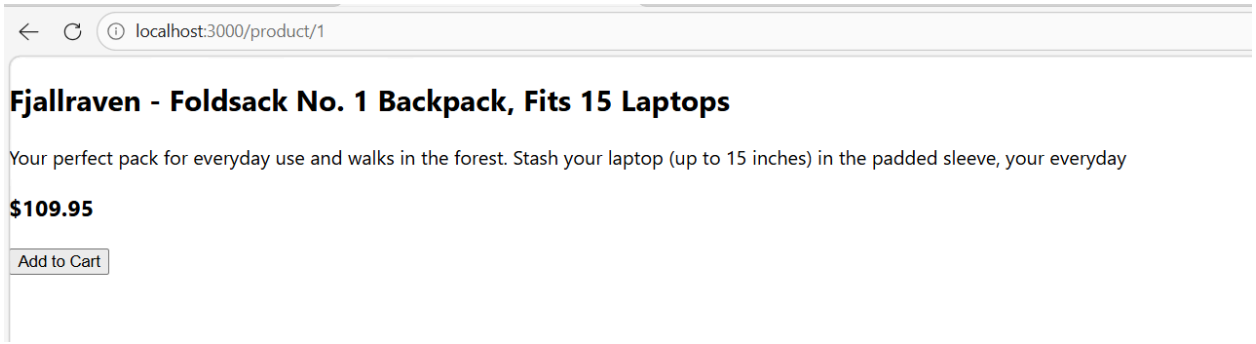
$168

White Gold Plated Princess

$9.99

Pierced Owl Rose Gold Plated Stainless Steel Double

$10.99



Beautify this:

Index.css

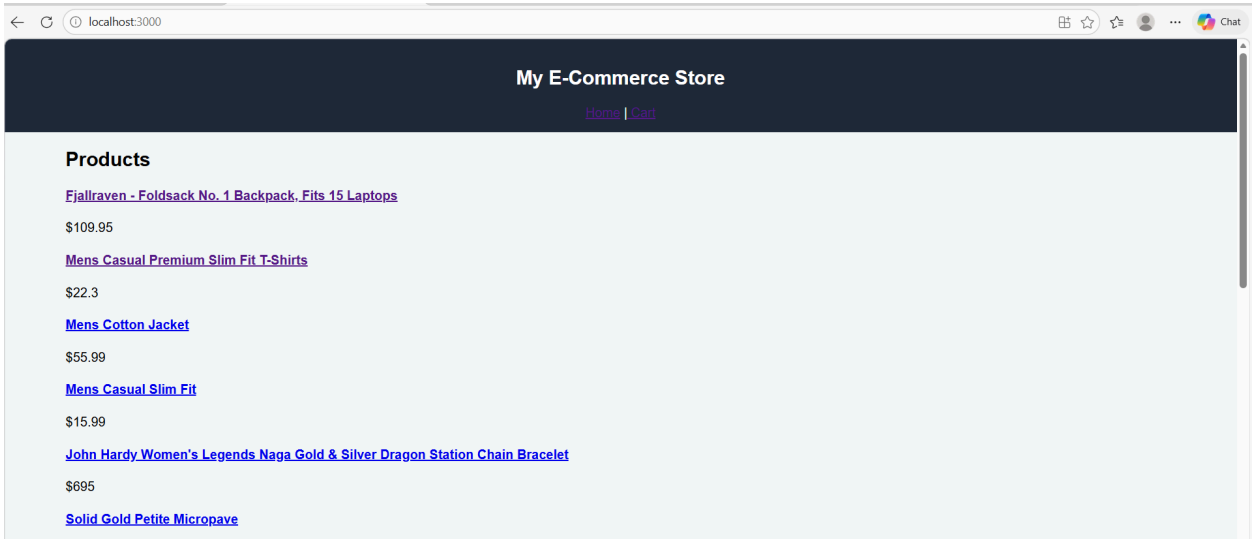
```
body {
  margin: 0;
  font-family: Arial, sans-serif;
  background-color: #f4f6f8;
}

.container {
  width: 90%;
  margin: auto;
}

.header {
  background: #1e293b;
  color: white;
  padding: 15px;
  text-align: center;
}

.product-grid {
  display: grid;
  grid-template-columns: repeat(auto-fill, minmax(220px, 1fr));
  gap: 20px;
  margin-top: 20px;
}

.product-card {
  background: white;
  padding: 15px;
```

```
  border-radius: 8px;
  box-shadow: 0px 2px 8px rgba(0,0,0,0.1);
  text-align: center;
}

.product-card img {
  height: 150px;
  object-fit: contain;
}

.product-card h4 {
  font-size: 16px;
}

.price {
  color: green;
  font-weight: bold;
}

.btn {
  background: #2563eb;
  color: white;
  border: none;
  padding: 8px 14px;
  margin-top: 8px;
  border-radius: 5px;
  cursor: pointer;
}

.btn:hover {
  background: #1d4ed8;
}
```

Create HEader - [APP.js]

```
import React from "react";
import { BrowserRouter, Routes, Route, Link } from "react-router-dom";

import ProductList from "./components/ProductList";
import ProductDetail from "./components/ProductDetail";
import Cart from "./components/Cart";
```

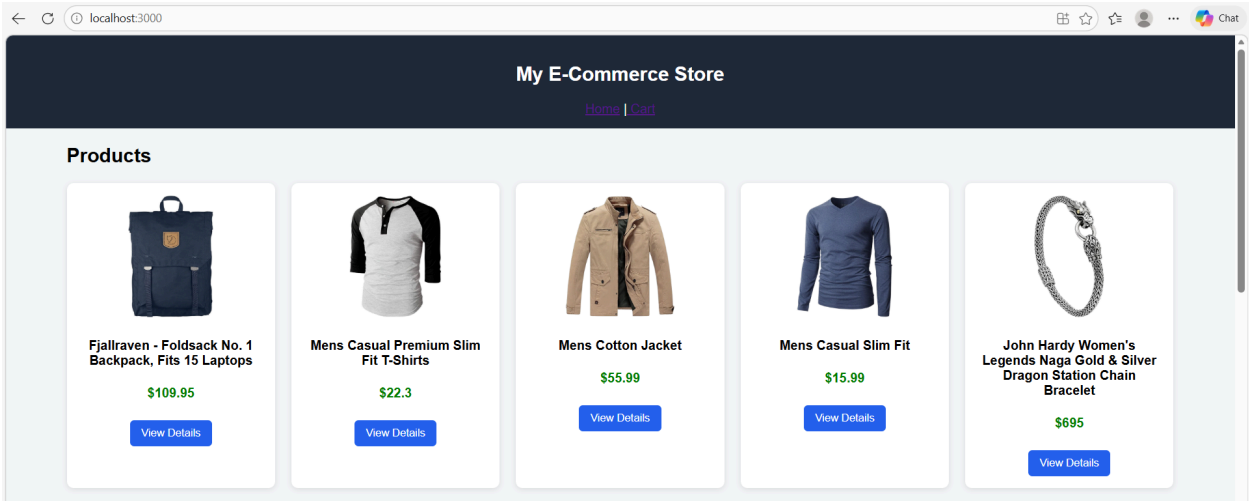
```
import Checkout from "./components/Checkout";

function App() {
  return (
    <BrowserRouter>

      <div className="header">
        <h2>My E-Commerce Store</h2>
        <Link to="/">Home</Link> |
        <Link to="/cart"> Cart</Link>
      </div>

      <div className="container">
        <Routes>
          <Route path="/" element={<ProductList />} />
          <Route path="/product/:id" element={<ProductDetail />} />
          <Route path="/cart" element={<Cart />} />
          <Route path="/checkout" element={<Checkout />} />
        </Routes>
      </div>

    </BrowserRouter>
  );
}

export default App;
```



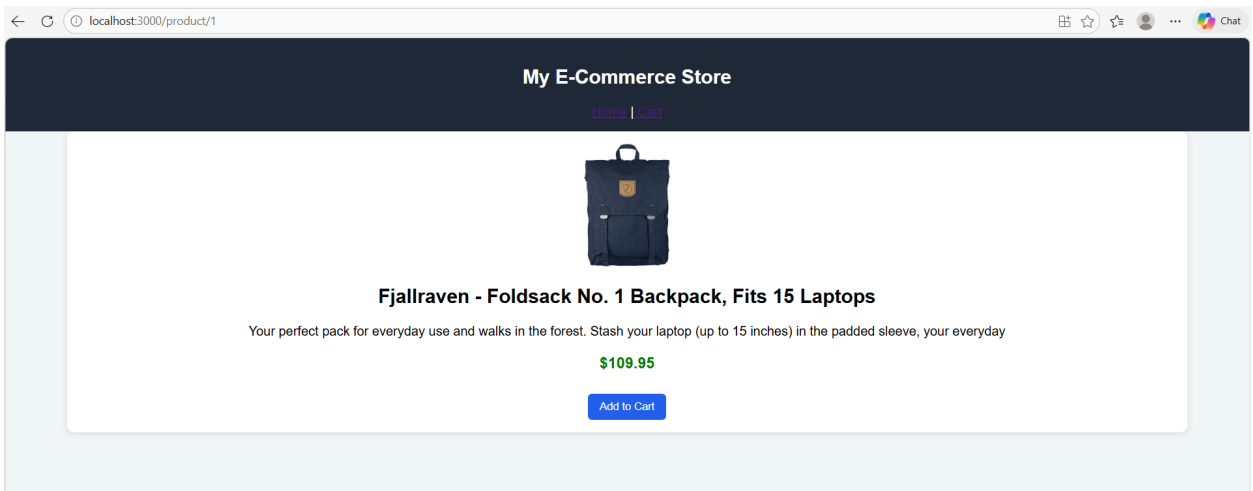
Improve product list ui

```
import React, { useEffect, useState } from "react";
import { getProducts } from "../services/api";
import { Link } from "react-router-dom";

function ProductList() {

  const [products, setProducts] = useState([]);

  useEffect(() => {
    getProducts().then(res => setProducts(res.data));
  }, []);

  return (
    <div>

      <h2>Products</h2>

      <div className="product-grid">

        {products.map(product => (
          <div className="product-card" key={product.id}>

            <img src={product.image} alt={product.title} />

            <h4>{product.title}</h4>

            <p className="price">${product.price}</p>

            <Link to={`/product/${product.id}`}>
              <button className="btn">View Details</button>
            </Link>

          </div>
        ))}

      </div>

    </div>
```

```
  );
}

export default ProductList;
```



Make Product Detail Look Better - [ProductDetail.js]

```
import React, { useEffect, useState } from "react";
import { useParams } from "react-router-dom";
import { getProduct } from "../services/api";

function ProductDetail() {

  const { id } = useParams();
  const [product, setProduct] = useState({});

  useEffect(() => {
    getProduct(id).then(res => setProduct(res.data));
  }, [id]);

  return (
    <div className="product-card">

      <img src={product.image} alt={product.title} />

      <h2>{product.title}</h2>
```

```
      <p>{product.description}</p>

      <h3 className="price">${product.price}</h3>

      <button className="btn">Add to Cart</button>

    </div>
  );
}

export default ProductDetail;
```